13 STATISTICS

Not a correct average

The monthly income of 10 households in a neighbourhood are these:

16500	21700	18600	21050	19500
17000	21000	18000	22000	17500

What is the mean monthly income?

Adding all these and dividing by 10, we get the mean monthly income as 19285 rupees.

Now, if instead of taking all these incomes separately, we had only the mean, then also we can make some conclusions about the general economic status of the households:

- The monthly income of all these households are around 19285 rupees.
- None of the households has a monthly income very much greater or very much less than 19285 rupees.
- The number of households with monthly income greater than 19285 rupees is more or less equal to the number of households with monthly income less than 19285 rupees.

Now suppose someone with a monthly income of 175000 rupees comes to live in the neighbourhood. What is the mean monthly income of the 11 households?

$$\frac{(19285 \times 10) + 175000}{11} \approx 33441 \text{ rupees.}$$

Without giving all these details, if this mean only is given, wouldn't we make the wrong conclusion that all these households have a monthly income around 30000 rupees? This is almost one and a half times the monthly income of ten of these households.

The purpose of calculating the mean is to reduce a whole collection of numbers to a single number, which gives a general understanding of a situation. But numbers in the collection which are very much less or very much more than others (though few) affect the mean a lot.

In our example, it was a single number very much larger than the first ten which changed the mean so much. Can you think of other instances like this where very small or very large numbers influence the mean to give a wrong impression?

Another average

Let's see how we can compute another average which gives a better overall indication of the monthly income of the 11 households. If we write all the incomes in increasing order and take the middle number, 5 of the households would have income less than this and 5 of them would have more.

Let's write the numbers in order:

16500, 17000, 17500, 18000, 18600, 19500,

21000, 21050, 21700, 22000, 175000

The middle number is 19500. It is called the *median* of these numbers. That is, the median monthly income of the 11 households is 19500 rupees. We can put it like this: of all the 11 households, 5 have monthly income less than 19500 rupees and 5 have more than 19500 rupees. That is, the number of households with income less than the median and the number of households with income more than the median are equal.

What if we take only the first 10 households? If we write incomes in the increasing order, there would be two numbers, 18600 and 19500 at the middle, instead of just one number.

Here also, we must choose the median such that the number of items below it and above it are equal. Any number between 18600 and 19500 would do for this. Usually half their sum is taken as the median.

That is, the median monthly income of the first 10 households is

$$\frac{1}{2}(18600 + 19500) = 19050 \text{ rupees}$$

The median income 19050 rupees, like the mean income 19285 rupees gives a reasonable estimate of the economic status of the first ten households (and there is no great difference between the mean and the median either).

What is important here is that the high income of the eleventh household does not change the median much. Also if we say that the median income of some households is 19050

rupees and that the monthly income of one of these is 21000 rupees, we can conclude that this household is better off than more than half the households considered.

(1) The distance covered by Ahirath in long jump practice are

6.10, 6.20, 6.18, 6.20, 6.25, 6.21, 6.15, 6.10

in metres. Find the mean and median. Why is it that there is not much difference between these?

(2) The table below gives the rainfall during the first week of June 2025 in various districts of Kerala.

District	Rainfall (mm)
Kasaragod	108.7
Kannur	89.4
Kozhikode	74.8
Wayanad	72.0
Malappuram	42.6
Palakkad	35.7
Thrissur	66.4
Ernakulam	73.5
Kottayam	69.1
Idukki	50.5
Pathanamthitta	43.6
Alapuzha	93.1
Kollam	39.0
Thiruvananthapuram	37.5

Calculate the mean and median rainfall in Kerala during this week. Why is the mean less than median ?

(3) Prove that for a set of numbers in arithmetic sequence, the mean and median are equal.

Frequency and median

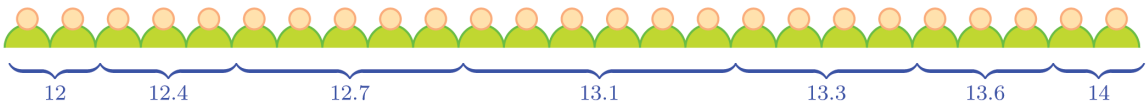
The amount of haemoglobin in blood is usually given as grams per decilitre (that is, 100 millilitres). The table below shows 25 children sorted according to haemoglobin levels, after a blood test.

Haemoglobin (g/dL)	Number of children
12.0	2
12.4	3
12.7	5
13.1	6
13.3	4
13.6	3
14.0	2

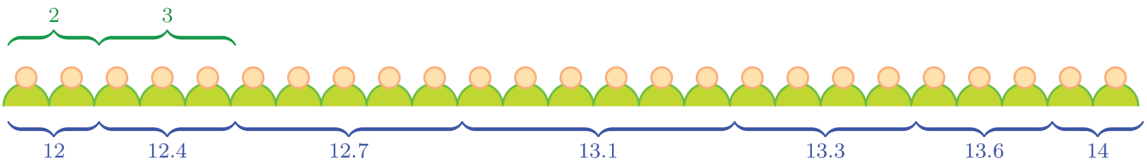
From this, we can compute the mean haemoglobin level. How do we find the median?

The median is that which comes in the middle; that is in this table of 25 children 12 of the children should have haemoglobin level less than the median level and 12 more than the median level.

To find it, we need only make the kids stand in a line, in the order of haemoglobin level and take the level of the thirteenth kid. Imagine the kids standing like this;



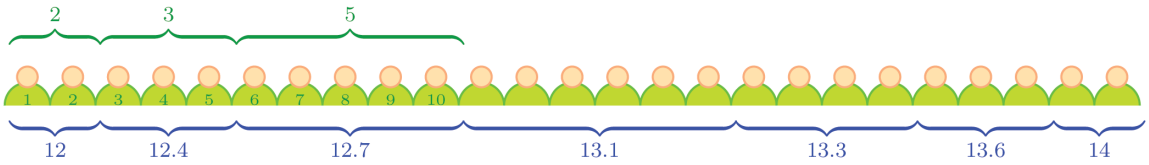
The first two have haemoglobin 12, the next 3 have 12.4 and thus the line grows.



We want the haemoglobin level of the 13th kid. By adding the numbers in the table one by one, we can find his position in the haemoglobin sequence.

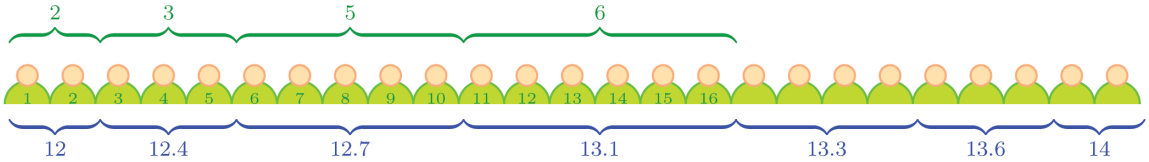
Taking the 2 + 3 = 5 kids of the first two groups, the level rises to 12.4. That is the 5th kid has level 12.4.

Adding the 5 kids in the next group, we have 5 + 5 = 10 kids and the level reaches 12.7.

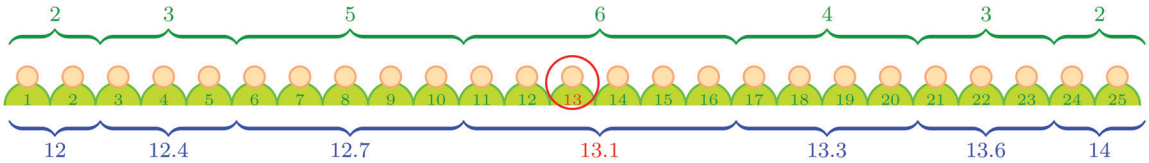


Thus the 10th kid is at level 12.7.

Adding the 6 kids in the next group, we have 10 + 6 = 16 kids.



We need the level of the 13th kid. The level of all the kids from the 11th to the 16th in the line is 13.1. So the 13th kid also has this level, and this is the median level.



We can do this using a table instead of a picture:

Haemoglobin (g/dL)	Number of children
up to 12.0	2
up to 12.4	5
up to 12.7	10
up to 13.1	16
up to 13.3	20
up to 13.6	23
up to 14.0	25

From the table, we can see the haemoglobin level of the kids, from the 11th to the 16th, is 13.1. Since the middle one of 25, that is the 13th, is in this set, the median level can be found as 13.1.

(1) 35 households in a neighbourhood are sorted according to their monthly income in the table below.

Monthly income (Rs)	Number of households
10000	3
11000	7
12000	8
13000	5
14000	5
15000	4
16000	3

Calculate the median income.

(2) The table below shows the workers in a factory sorted according to their daily wages:

Daily wages (Rs)	Number of workers
700	2
800	4
900	5
1000	7
1100	5
1200	4
1300	3

Calculate the median daily wage.

(3) The table below gives the number of babies born in a hospital during a week, sorted according to their birth weight.

Weight (kg)	Number of babies
2.500	4
2.600	6
2.750	8
2.800	10
3.000	12
3.150	10
3.250	8
3.300	7
3.500	5

Calculate the median birth-weight.

Classes and median

The table below shows the workers of a factory sorted according to their wages:

Daily Wages (Rs)	Number of Workers
400 - 500	6
500 - 600	7
600 - 700	10
700 - 800	9
800 - 900	5
900 - 1000	4
Total	41

How do we compute the median daily wage in this factory?

What we want to calculate is the daily wage of the worker in the middle, when the workers are arranged in order from the one earning the least to the one earning the most. Here there are 41 workers in all; so the one in the middle is the 21st in this order.

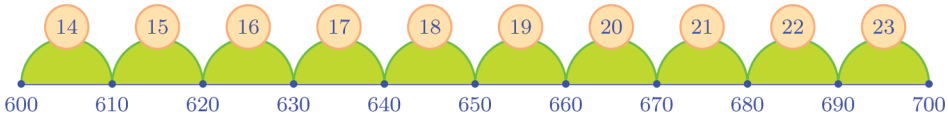
The table splits the wages into different classes. Let's first find the class to which the 21st person belongs. As in our previous problems, let's calculate the total when each class is joined to all classes before it:

Daily Wages (Rs)	Number of Workers
Below 500	6
Below 600	13
Below 700	23
Below 800	32
Below 900	37
Below 1000	41

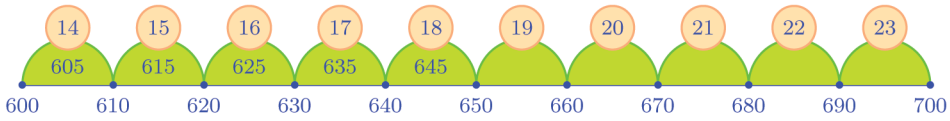
We see from the table that when we take together all those earning less than 600 rupees a day, then we reach up to the 13th person; and when we take together all those earning less than 700 rupees a day, then we reach up to the 23rd person. The 21st person we seek is between them. Thus we find that his daily wage is between 600 and 700 rupees.

What do we do to make this more exact? We only know that the 10 workers from the 14th to the 23rd earn between 600 and 700 rupees; we don't know what the actual earning of each is.

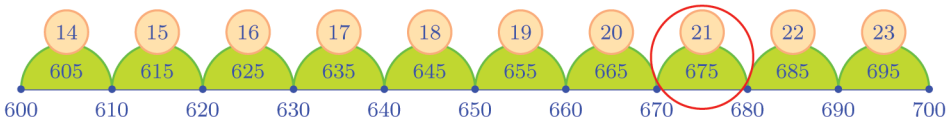
So, we need to make some assumptions. (We made some assumptions when we calculated the mean from a table involving classes and frequencies, remember?). We divide the 100 rupees from 600 to 700 into 10 equal parts and assume that each of these subdivisions contains exactly one person:



We further suppose that the daily wage of each of these workers is the midvalue of the class to which he belongs:



Under these assumptions, we can compute the daily wage of the 21st worker:



We can calculate this without drawing pictures. How did we compute the median wage, on the basis of our assumptions regarding the arrangement of workers?

- The daily wage of the 14th worker is 605 rupees.
- The daily wage of each one thereafter, till the 23rd increases by 10 rupees.
- To reach the 21st worker we want, from the 14th, we must take 7 more workers.

Now it is an arithmetic problem, isn't it?

What number do we get if we start from 605 and add 10 repeatedly 7 times?

We can calculate it as

$$605 + (7 \times 10) = 675$$

Haven't we seen many such problems in the chapter, Arithmetic Sequences?

Let's do one such problem, without drawing pictures. The table below shows the employees in an office, sorted according to their age:

Age	Number of Workers
25 - 30	4
30 - 35	7
35 - 40	8
40 - 45	10
45 - 50	9
50 - 55	8
Total	**46**

We want to compute the median age. Here the total number is 46, which is an even number. So we arrange the employees from the youngest to the oldest and take half the sum of the ages of the 23rd and the 24th persons as the median age. First let's write the cumulated frequencies:

Age	Number of Workers
Below 30	4
Below 35	11
Below 40	19
Below 45	29
Below 50	38
Below 55	46

According to this, the ages of the 10 persons from the 20th to the 29th positions in the order of ages, are between 40 and 45. The persons in the 23rd and 24th position whom we want are in this group.

As in the previous problem, we divide the 5 years between 40 and 45 into 10 equal parts, and assume that each such subdivision contains one person whose age is the midvalue of his class.

Then each subdivision is $\frac{5}{10}=\frac{1}{2}$ year. The age of the 20th person is the midvalue of 40 and $40\frac{1}{2}$, which is $40\frac{1}{4}$. Our assumption means the age of each succeeding person increases by $\frac{1}{2}$ year (till the 29th). So the age of the 23rd person is

$$40\frac{1}{4}+\left(3\times\frac{1}{2}\right)=40\frac{1}{4}+1\frac{1}{2}=41\frac{3}{4} \text{ years}$$

and the age of the 24th person is

$$41\frac{3}{4}+\frac{1}{2}=42\frac{1}{4} \text{ years}$$

Now to get the median age, we take half the sum of these two:

$$\frac{1}{2}\left(41\frac{3}{4}+42\frac{1}{4}\right)=\frac{1}{2}\times 84=42$$

So the median age is 42. In this example, the persons in the 23rd and 24th positions are in the same class 40 – 45. In contexts where this is not so, we compute the median in a slightly different manner. See this problem.

The table below shows children of a class sorted according to their marks in an exam:

Marks	Number of Children
0 - 10	4
10 - 20	7
20 - 30	9
30 - 40	12
40 - 50	8
Total	**40**

We want to compute the median mark.

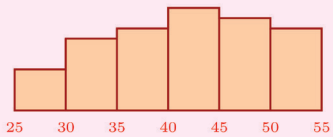
First let's write the cumulated frequencies:

Marks	Number of Children
Below 10	4
Below 20	11
Below 30	20
Below 40	32
Below 50	40

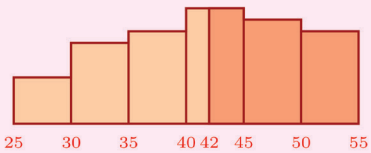
Here, if we arrange the children from the one with the least mark to the one with the greatest, then the number of children with marks below 30 is 20, which is exactly half the total number of 40 children.

Median area

Do you remember drawing histograms of frequency tables? The histogram of the age problem is like this:



In this, the vertical line through the median splits the picture into two parts:



It is not difficult to see that the areas of the two parts are equal (try it!)

Does the median have this property in all such computations?

In this case, we take 30 itself as the median mark. The justification for this is that half the total number of children have marks below 30 and half of them have marks 30 or more.

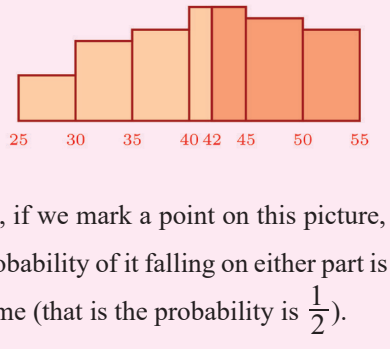
(1) The table shows some households sorted according to their usage of electricity:

Electricity usage (units)	Number of households
220 - 240	3
240 - 260	6
260 - 280	7
280 - 300	10
300 - 320	9
320 - 340	4

Calculate the median usage of electricity.

Median probability

We have noted that the perpendicular through the median divides the histogram into two parts of equal area:



So, if we mark a point on this picture, the probability of it falling on either part is the same (that is the probability is $\frac{1}{2}$).

This means that in the office, given in the problem, if we select someone without any special consideration, the probability of his age being less than 42 or more than 42 is the same.

(2) The table below shows the children in a class sorted according to their marks in the math exam:

Marks	No. of Children
0 - 10	4
10 - 20	8
20 - 30	10
30 - 40	9
40 - 50	5

Calculate the median mark of the class.

(3) The table below gives the details of the income tax paid by the employees in an office in a year:

Income tax (Rs)	Number of employees
1000 - 2000	8
2000 - 3000	10
3000 - 4000	15
4000 - 5000	20
5000 - 6000	22
6000 - 7000	8
7000 - 8000	6
8000 - 9000	3

Calculate the median income tax paid.